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1.0 Overview of The Project

The digital transformation of service-based industries has significantly accelerated the demand for sophisticated, technology-driven platforms. These platforms are essential for managing increasingly complex transactional processes with both efficiency and adaptability. As the nature of service delivery evolves, there is a growing expectation that technology will not only streamline operations but also respond dynamically to diverse and changing user needs. Consequently, the development and implementation of intelligent systems that can seamlessly integrate various service components have become critical to maintaining competitive advantage and ensuring high-quality customer experiences in this rapidly changing landscape.

Within the domain of property rentals, traditional systems have exhibited notable limitations, particularly in addressing the diverse and constantly evolving scheduling demands of modern users. These platforms often lack the flexibility required to accommodate the complex and dynamic nature of current rental arrangements, thereby restricting their effectiveness in meeting user expectations for adaptability and convenience. These legacy platforms often operate on rigid, predefined booking structures that fail to address the specific and changing time-related (temporal) and situational (contextual) requirements of both service providers and service renters. The operations need to rely entirely on a manual third-party messaging and cross-referencing.

FlexRent has been introduced as a property rental system designed with AI features and limitless flexibility bookings for users based on hourly, daily and monthly durations. The system enables users to function as both property seekers and providers, facilitating a seamless and intelligent rental experience. It achieves this by leveraging AI-driven recommendations that use user-provided keywords to identify and address their specific needs accurately. Moreover, we included an AI algorithm that recommends users based on past bookings and searches.

2.0 Problem Statement

Despite advancements in digital marketplace platforms, existing systems remain limited in flexibility, personalization, and efficiency. Users encounter several issues, including:

- Users struggle to find suitable options efficiently due to limited personalization
- Users have to manually cross-reference their physical notebooks or personal digital calendars before replying.
- Users have to perform an online transfer and make manual transactions via WhatsApp.
- Booking conflicts occur frequently due to poor time-slot management.
- Manual booking modifications, expecting custom penalty rules, collect the user's details and execute a return transfer for different rental times, such as daily and monthly.
- Dependence on multiple platforms to fulfil different requirements.

These limitations result in reduced user satisfaction and inefficient resource utilization. Therefore, there is a need for an AI-integrated platform that can provide flexible booking options, intelligent recommendations, and efficient scheduling. Furthermore, FlexRent provides an all-in-one solution for different rental periods.

3.0 Proposed Solution

The proposed solution is the development of the FlexRent AI Marketplace, which integrates advanced features to address the identified problems.

3.1 Technical Feasibility

The system can be developed using widely available and reliable technologies:

- Frontend: Python
- Backend: Streamlit
- Database: SQLite
- AI Techniques: Rule-based algorithm

The system will be developed using Python with the Streamlit framework, which allows rapid development of web-based applications without requiring separate frontend and backend implementations. SQLite will be used as the database due to its simplicity and ease of integration. AI functionalities will be implemented using rule-based algorithms to provide basic recommendations. This approach ensures faster development, reduced complexity, and suitability for project implementation.

3.2 Operational Feasibility

The system is designed to be user-friendly and efficient:

- Provides an intuitive interface for both seekers and providers
- Reduces manual effort through automation
- Improves booking efficiency with smart scheduling
- Enhances user satisfaction through personalized recommendations

Thus, the system is practical and can be effectively implemented in real-world environments.

3.3 Economic Feasibility

The benefits significantly outweigh the costs, making the project economically reliable for the system development:

Assumptions	
Discount rate	10%
Annual increment (cost)	10%
Annual increment (benefit)	4%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefit)	0.9

Estimated cost	
Development	
Hardware	15000
Software	30000
Testing and deployment	5000
Production	
Server and cloud hosting	3600
Domain and security	400
Marketing and ads	12000
Maintenance	6000
Estimated benefits	
Commission fee	60000
Listing fees	5000
Penalty fees	3000
Advertising fees	10000

A. Development cost	Year						
	0	1	2	3	4	5	6
Hardware	16500	-	-	-	-	-	-
Software	33000	-	-	-	-	-	-
Testing and deployment	5500	-	-	-	-	-	-
Total	55000	-	-	-	-	-	-
B. Production cost	Year						
	0	1	2	3	4	5	6

4.0 Information Gathering Process

The system is implemented based on the criteria needed by users to create mandatory features in the next phase, which is the TO-BE Logical DFD. In this information-gathering process, we conducted a survey to analyse the requirements of FlexRent users and rental providers for developing an improvement to the rental platform. The method and a summary of the survey are detailed in 4.1 and 4.2.

4.1 Method Used

The current analytics of the FlexRent platform employ a questionnaire approach to thoughtfully gather detailed information about the needs and preferences of both users and rental providers. This method allows for a clear understanding of their requirements, helping to ensure the platform meets their expectations effectively.

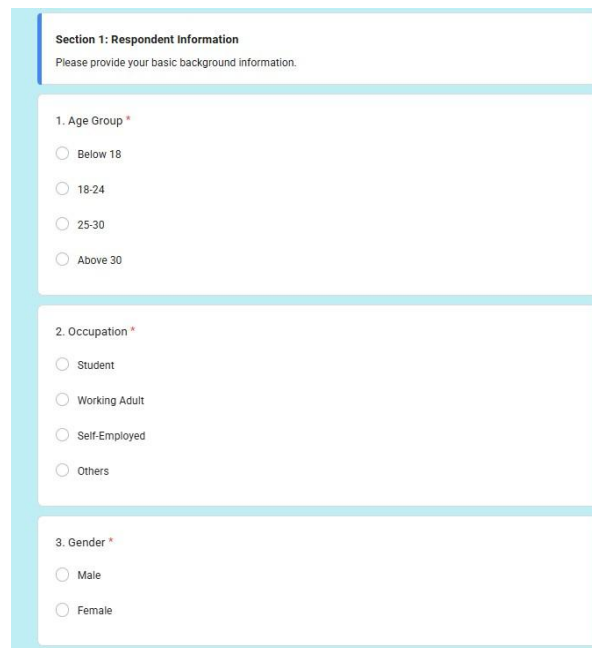


The screenshot shows a Google Form titled "FlexRent: A Property Rental System Marketplace Platform Survey". The form includes a header with the title and a sub-header "FlexRent: A Property Super App". Below the header, there is a paragraph stating: "We are The Analyst Group, developing a project titled 'FlexRent: A Property Super App', which focuses on property rental solutions." The form also lists the group members: 1. Mohamad Azryq Haiqal Bin Mohd Rosli, 2. Bazli Abdullah Bin Mohd Yusof, 3. Muhamad Afif Azhar Bin Mohd Shahrizan, 4. Mikhail Adam Riaz Bin Yazmi, and 5. Ahnaf Bin Zaizal. The survey objectives are listed as: To analyze user experiences with the current property rental system, To determine user requirements and proposed features for an improved property rental platform (FlexRent), and To identify problems and limitations faced by users in existing systems. The form also states that the survey contains 15 questions: 12 close-ended questions and 3 open-ended questions.

Figure 4. 1: FlexRent Google Form Survey

In Figure 4.1, we created a Google Form survey that consists of a survey title and objectives. The survey is mainly focusing on the rental platform requirements. The questionnaire consists of a total of 15 questions, with 12 being closed-ended to gather specific responses, and 3 open-ended questions designed to capture more detailed and personalized feedback. To ensure clarity and smooth flow, the questions will be arranged into four sections, each addressing a different area of the survey.

4.2 Summary from method used (Interview/Questionnaire/Observation)

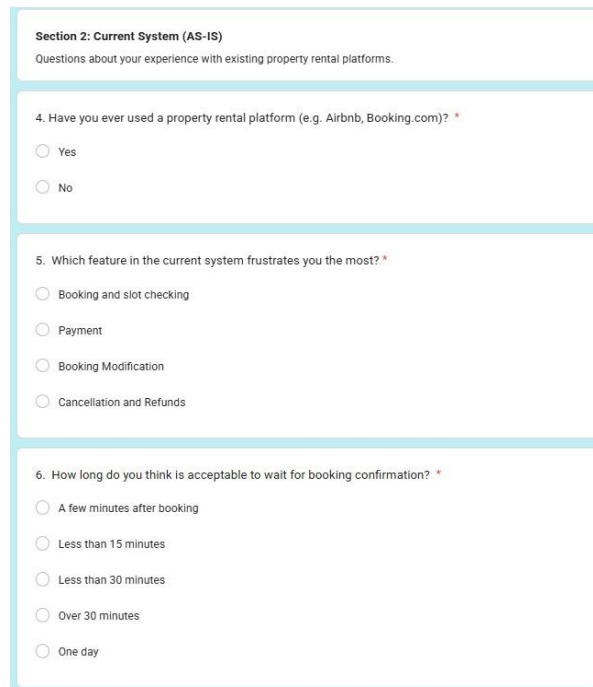


The form is titled "Section 1: Respondent Information" and includes the instruction "Please provide your basic background information." It contains three sections of radio button questions:

- 1. Age Group ***
 - ☐ Below 18
 - ☐ 18-24
 - ☐ 25-30
 - ☐ Above 30
- 2. Occupation ***
 - ☐ Student
 - ☐ Working Adult
 - ☐ Self-Employed
 - ☐ Others
- 3. Gender ***
 - ☐ Male
 - ☐ Female

Figure 4. 2: Respondent Information

In Figure 4.2, we prepared a section for the respondent to justify their personal information. These proposed questions are to collect the comparison between our colony based on their exposure to a rental platform.



The form is titled "Section 2: Current System (AS-IS)" and includes the instruction "Questions about your experience with existing property rental platforms." It contains three sections of radio button questions:

- 4. Have you ever used a property rental platform (e.g. Airbnb, Booking.com)? ***
 - ☐ Yes
 - ☐ No
- 5. Which feature in the current system frustrates you the most? ***
 - ☐ Booking and slot checking
 - ☐ Payment
 - ☐ Booking Modification
 - ☐ Cancellation and Refunds
- 6. How long do you think is acceptable to wait for booking confirmation? ***
 - ☐ A few minutes after booking
 - ☐ Less than 15 minutes
 - ☐ Less than 30 minutes
 - ☐ Over 30 minutes
 - ☐ One day

Figure 4. 3: Current System (AS-IS)

The AS-IS of the current system section defines the most significant problems users face based on their previous activity on the rental platform. Figure 4.3 shows the elements of questions being asked to the respondent as part of gathering information for the FlexRent platform.

Section 3: Proposed System (to-be)
Questions about the features and improvements you would like to see in FlexRent System.

9. Would you prefer a platform that combines multiple rental services in one place? *

☐ Yes

☐ No

10. Which features are important to you in a property rental platform? *

☐ Flexible schedule system (hourly, daily, monthly, and yearly rental)

☐ Search and filtering options

☐ Smart Matching

☐ Automatic payment modification

☐ AI Recommendations

11. How important is flexible booking to you? *

Not important - 1

Very Important - 5

12

1
2
3
4
5

☆
☆
☆
☆
☆

Figure 4. 4: Proposed System (To-Be)

This section is about the system features and improvements which are needed in the proposed system. The proposed system has to fulfil the users and rental providers requirements to establish a well-performing rental platform.

Section 4: Open-Ended Questions
Please share your thoughts in the spaces provided below.

13. What problems do you usually face when using a property rental platform? *

Long answer text

14. What improvements would you like to see in a rental platform? *

Long answer text

15. Any additional suggestions or comments? *

Long answer text

Figure 4. 5: Open-Ended Questions

In Figure 4.5, the respondents will need to give their feedback according to the previous activities of using a rental platform. This section is to get the respondent's idea on what they want to see in future for a rental platform.

5.0 Requirement Analysis (based on AS-IS analysis)

The Requirement Analysis was developed through structured stakeholder engagement, and we produced a clear articulation of both functional requirements. These cover the core platform features across event space bookings, short-term stays, and long-term leases. The non-functional requirements address critical concerns such as security, payment integrity, and AI performance metrics. To complement this, a Logical DFD package was constructed, capturing the AS-IS state of the system through a Context Diagram and Level 0 Data Flow Diagram that traces how information currently moves across a manual WhatsApp- based workflow.

This diagram reveals a process heavily dependent on human intervention at every stage, from inquiry handling and availability checks to payment confirmation and tenancy documentation. Without any automated layer to validate, route, or record these interactions reliably. Both deliverables establish a grounded baseline of where the current system falls short, forming the analytical foundation from which the proposed FlexRent platform requirements are derived and justified.

5.1 Current business process (scenarios, workflow)

Listing Discovery: Users search for desired rental properties and are shown a listed catalogue, then request slot availability through a directory via WhatsApp.

Manual Availability Check: The Provider opens the chat, reads the requested dates/hours, and opens a physical notebook to manually check availability.

Manual Bank Transfer Processing: The Provider texts back their bank account number. The User opens their online banking application, executes the transaction, captures a screenshot of the receipt, and uploads it to the chat.

Manual Booking Verification: The Providers log in to their bank portal, scan their transaction history to confirm receipt of the funds, and manually log the reservation into their notebook.

Manual Adjustments/Cancellations: If a change is needed, the User messages the Provider. The Provider manually gives out the refund if there are cancellations and updates booking slots in the notebook.

5.2 Functional Requirement (input, process and output)

5.2.1 Accounts Registration and Identity

Input:

- User registration details: Name, Email, Password, Role Choices, and Phone Number.
- Login credentials and profile setup requirements.

Process:

- Authentication of the user's credentials, followed by validation of the user's information by the system.

Output:

- Active verified account session, confirmation emails, and updated profile configurations.
- The system also stores all the data and information securely and can be retrieved for future use.

5.2.2 Property Listing Management (Search, Filter, and Recommendation)**Input:**

- Included property details, images, pricing, rental duration, and availability.
- Textual search criteria such as keywords, location, date, and budget.
- Algorithm from past user booking and search history.

Process:

- The system stores and updates listing information for providers.
- The system filters the listing based on the user's need and stored property data.

Output:

- Displayed tailored personalized property listings.
- Updated rental listings provided to seekers.

5.2.3 Booking and Availability (Real-Time)**Input:**

- Desired rental time slot selection by seekers: Hourly, Daily, Weekly, or Monthly.
- Selected property, booking duration of date and time, and payment confirmation.

Process:

- The system automatically checks the real-time calendar and validates the property's availability slot.
- The system reserves the booking slot followed by a booking transaction.

Output:

- Displayed the availability of the chosen property by seekers.
- Confirmed booking notification updated the real-time calendar.

5.2.4 Payment Gateway and Invoicing**Input:**

- Credit or Debit card details or online banking credentials.
- Booking transaction totals.

Process:

- The system processes digital payments through an integrated payment service provided.
- Calculation of total transaction fees, including applicable platform commission, discount, or listing fees.

Output:

- Payment confirmation notification.
- Displayed downloadable automated digital receipt and invoice.
- Updated booking status confirmed as “Paid”.

5.2.5 Modification and Cancellation Management**Input:**

- Booking and refund policy data for both parties.
- Self-service request (rescheduling or cancellation) notices submitted by seekers.

Process:

- Query of existing booking records provided against the system cancellation policy.
- Calculation of penalty fees and system refunds automatically.
- The system frees up the cancelled slot.

Output:

- Rescheduling booking confirmation alert (if needed).
- Cancellation verification.
- Updated real-time booking calendar.
- Provided refund receipt logs.

5.3 Non-functional Requirement (performance and control)

- **Security Systems**

The platform must implement very strict visual and operational functions between seekers and providers under the role-based access management. Every login session must be followed by user authentication using an encrypted password. It also must be able to secure the data and sensitive user information collected during registration and profile setup. The privacy information regarding booking histories and payment records must be stored securely in a database and accessible only to authorised users.

Furthermore, production systems must utilize secure configurations and be backed up by trusted domain certificates with system-wide security frameworks to prevent unauthorized access.

- **Payment Integrity**

The system must provide an integrated payment gateway that executes real-time confirmation checks, outsourcing the need for manual receipt checking via manually verifying bank portal lines. The system also needs to provide financial traceability for every payment transaction, platform commission fee, listing fees, and penalty charges that could be tracked with a matching digital receipt log mapped directly towards the database. In addition, all transactions that occurred

have to be validated automatically to prevent fraud or transaction-mismatch errors.

For accuracy, the system could provide automated custom penalty calculators that could resolve balance discrepancies instantly when dealing with rescheduling or cancellation by seekers, thus eliminating human estimation mistakes.

- **AI Performance Metrics**

AI performance as required by this system must be maintained thoroughly from time to time to ensure the smoothness and the best user experience. The standardization of AI started from recommendation latency when it helps seekers searching and filtering based on algorithms of past booking history and keywords to deliver tailored listings on the search dashboard with minimal overhead to preserve its responsiveness. The rule-based algorithms also must accurately analyse user-provided keywords to return highly compatible options appropriate to the user's needs.

The AI must be able to match diverse data provided in time demands, duration of stay, and budget provided by users. The logical engines of AI must be assured of their consistency uptime alongside the frontend and database layers to continuously automate the filtering process without the need of manually cross-referencing fallbacks.

5.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)

- **Context Diagram**

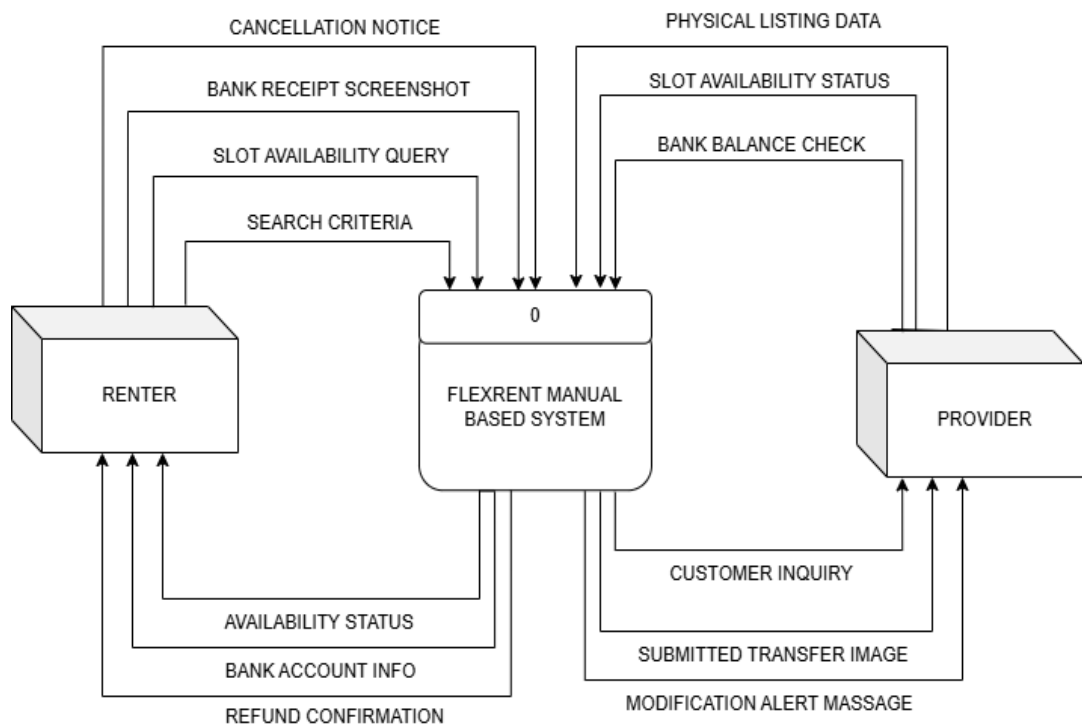


Figure 5. 1: Current FlexRent Context Diagram

A context diagram, often referred to as a Level-0 Data Flow Diagram (DFD), represents the most abstract level of system visualization. It depicts the system as a singular process, or "black box," and identifies all external entities that interact with the system. The connections between the system and these entities illustrate the flow of data, indicating the exchange of information entering and leaving the system.

- The Renter represents the customer segment within the business model. Their primary interaction with the system involves requesting and confirming information. When a renter intends to reserve a space, they initiate a manual search or inquire about slot availability. These interactions are deliberate human actions, typically conducted via WhatsApp or telephone, rather than automated website clicks. Renters also submit bank receipt screenshots as payment verification and may issue cancellation notices if their plans change.
- In response, the system provides the renter with essential information to facilitate the process, including a WhatsApp communication link, the current availability status of the listing, bank account details for payment, and confirmation of refunds upon approved cancellations.
- The Provider refers to the owner of the property or listing. Their role is to furnish the system with the essential data required for its operation. This includes physical listing information, manual calendar records indicating booking statuses, and configurable settings such as penalty rules for late cancellations or damages, as well as bank balance verifications, potentially used to confirm incoming payments.
- In return, the system keeps the Provider informed by transmitting customer inquiries, payment evidence in the form of transfer images submitted by renters, and notifications regarding any changes to bookings.

- **Diagram 0**

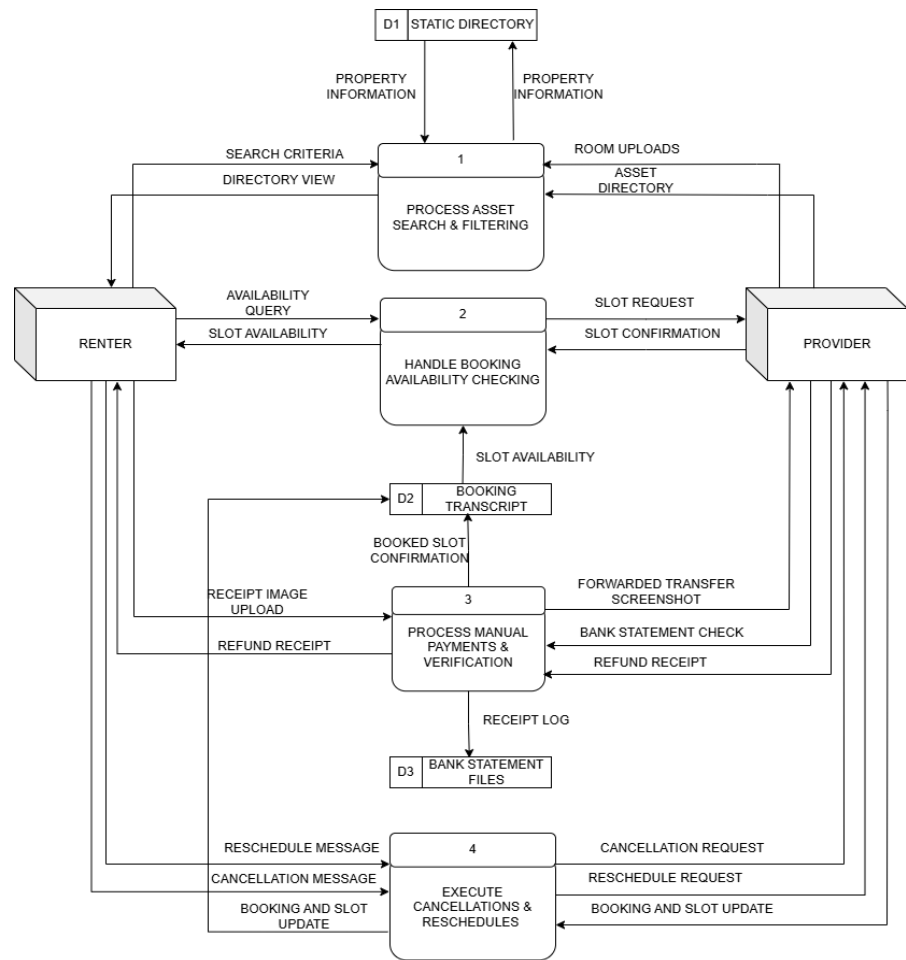


Figure 5.2: FlexRent Diagram 0

Diagram 0 of the FlexRent Manual-Based System is a Level-0 Data Flow Diagram (DFD) that decomposes the single Process 0 from the Context Diagram into four distinct yet interconnected sub-processes, revealing the internal operational logic of the rental platform across three data stores and two external entities, such as the Renter and the Provider.

- The user journey commences with Process 1, termed Asset Search & Filtering, where a Renter initiates interaction by submitting a search criteria request. This action prompts the system to query the D1 Static Directory, a manually maintained master list of property listings populated by providers through room uploads and asset directory submissions. Subsequently, the system returns a filtered view of the directory to the Renter, thereby functioning as the system's discovery and browsing interface.
- After identifying a listing of interest, the Renter's interaction moves to Process 2, Handle Booking Availability Checking. This process handles the manual coordination of time slots by receiving an availability request from the Renter, sending a slot inquiry to the Provider, and waiting for a confirmation. Once confirmed, the system relays the availability status to the Renter and

simultaneously records the confirmed slot in D2, the Booking Transcript, which acts as the system's short-term memory, bridging the booking and payment phases.

- This connection becomes essential in Process 3, Process Manual Payments & Verification, which references the booked slot confirmation from D2 to provide context for incoming payment evidence. This evidence includes the receipt image uploaded by the Renter, as well as the transfer screenshot and bank statement verification provided by the Provider. Upon verifying the transaction, the system issues a refund receipt to the Renter when applicable and permanently records the financial transaction in D3, the Bank Statement Files data store, thereby creating an audit trail for all monetary activities.
- Finally, when changes occur after a booking has been confirmed, Process 4, Execute Cancellations & Reschedules, manages post-booking lifecycle events. It receives cancellation and reschedule requests, along with booking slot updates from the Provider, and communicates the corresponding reschedule notifications, cancellation confirmations, and updated booking records back to the Renter, thus completing the operational cycle of the system.

6.0 Summary of Requirement Analysis Process

The requirements analysis of the existing AS-IS system for the FlexRent rental platform indicates a wholly manual, human-dependent operation encompassing four primary processes: asset search and filtering, booking availability checking, manual payment verification, and cancellation or reschedule execution. Each of these processes relies predominantly on direct human intervention rather than automated mechanisms, resulting in a system that, while operationally purposeful, remains vulnerable and fragile in its execution.

From the analysis of the business process, the functional requirements of the system include several key capabilities. The system must enable property listing management by allowing Providers to register and update rental assets within the directory. It must support rental search and filtering so that Renters can find available properties according to their specified criteria. The system should manage booking availability checking by coordinating slot requests and confirmations between Renters and Providers while recording the outcomes. It must also handle payment verification by accepting receipt image uploads from Renters, forwarding transfer evidence to Providers, performing bank statement cross-checks, and issuing refund confirmations when appropriate. The system needs to maintain accurate records in the Static Directory, Booking Transcript, and Bank Statement Files to preserve a reliable history of listings, bookings, and financial transactions. Finally, it must process post-booking modifications by managing cancellation and reschedule requests from both Renters and Providers and notifying them of the resulting changes.

In addition to these functional requirements, the non-functional requirements derived from the AS-IS analysis highlight several performance and quality expectations that the current system consistently fails to satisfy. Taken together, the observations of the current business processes, along with the identified functional and non-functional requirements, constitute the comprehensive requirements baseline for the AS-IS FlexRent system. This analysis reveals a platform that, while operationally coherent, is structurally limited. It also provides the analytical foundation for designing a proposed TO-BE system in the next phase aimed at replacing manual interventions with structured, automated, and scalable digital workflows.